

Chapter 8

The Local Obidi Action (LOA)

Entropy, Geometry, and the Variational Principle

8.1 Introduction

The preceding chapters established the geometric architecture of the Theory of Entropicity through the Hybrid Manifold Architecture System (HMAS), the Obidi Convention, and Obidi Calculus.

A physical theory, however, requires more than geometry and algebra.

It requires dynamics.

The dynamics of modern theoretical physics are generally formulated through an action principle.

In classical mechanics, the action generates Newton's laws.

In electromagnetism, the action generates Maxwell's equations.

In General Relativity, the Einstein–Hilbert action generates Einstein's field equations.

The Theory of Entropicity follows this tradition.

Its dynamical foundation is the **Local Obidi Action (LOA)**.

The LOA is the central variational principle from which all field equations of the theory are derived.

It is the mathematical object that connects:

- entropy,
- information geometry,
- spacetime geometry,
- gravitation,

within a unified framework.

8.2 The Principle of Least Action

The action principle occupies a privileged position in modern physics.

For a dynamical system with Lagrangian density

$$\mathcal{L},$$

the action is

$$I = \int \mathcal{L} dV.$$

The physical evolution of the system is determined by requiring that

$$\delta I = 0.$$

This condition states that the actual trajectory of the system is a stationary point of the action functional.

All field equations follow from this variational requirement.

The Theory of Entropicity adopts the same principle.

8.3 The Einstein–Hilbert Action

To understand the motivation behind the Local Obidi Action, it is useful to review the Einstein–Hilbert Action.

General Relativity is generated by

$$I_{EH} = \frac{1}{16\pi G} \int (R - 2\Lambda) \sqrt{|g|} d^4x.$$

Variation with respect to the metric yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

The Einstein–Hilbert Action therefore provides the dynamical foundation of spacetime geometry.

The Theory of Entropicity seeks to generalize this construction.

8.4 Why a New Action Is Needed

The Einstein–Hilbert Action assumes:

1. A single geometric sector.
2. A single metric.
3. No explicit entropy field.

These assumptions are incompatible with HMAS.

The Theory of Entropicity contains:

$$FR, \quad FS, \quad L.$$

Each sector possesses:

- its own metric,
- its own curvature,
- its own geometric dynamics.

Furthermore, entropy itself is treated as a dynamical field.

A generalized action is therefore required.

8.5 Entropy as a Dynamical Variable

The fundamental innovation of ToE is the elevation of entropy from a statistical descriptor to a dynamical field.

The entropy field is denoted

$$S(x).$$

Unlike conventional thermodynamics, where entropy merely characterizes a state, ToE treats entropy as an active participant in physical evolution.

Consequently the action must depend explicitly upon:

$$S(x),$$

and

$$\nabla_\mu S.$$

This requirement introduces entirely new dynamical possibilities.

8.6 Sectoral Action Principle

Each sector possesses its own local action contribution.

The sectoral action is defined as

$$I_{LOA}^{(J)}.$$

The total action is obtained through the Obidi Addition Rule:

$$I_{LOA} = \sum_J I_{LOA}^{(J)}.$$

Explicitly,

$$I_{LOA} = I_{FR} + I_{FS} + I_L.$$

This decomposition reflects the multi-sector structure of HMAS.

8.7 Construction of the Local Obidi Action

For a given sector J , the Local Obidi Action is defined by

$$I_{LOA}^{(J)} = \int \mathcal{L}^{(J)} dV^{(J)}.$$

The corresponding Lagrangian density is

$$\mathcal{L}^{(J)} = \left(R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu^{(J)} S \nabla_\nu^{(J)} S + V(S_{(J)}) \right) e^{S/k_B}.$$

The invariant measure is

$$dV^{(J)} = \sqrt{|g^{(J)}|} d^4x.$$

Combining these expressions yields

$$I_{LOA}^{(J)} = \int \left(R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu^{(J)} S \nabla_\nu^{(J)} S + V(S_{(J)}) \right) e^{S/k_B} \sqrt{|g^{(J)}|} d^4x.$$

This is the fundamental action of the Theory of Entropicity.

8.8 Interpretation of the Action Terms

Each component possesses a specific physical interpretation.

Curvature Term

$$R^{(J)}$$

represents geometric curvature.

It generalizes the Ricci scalar appearing in General Relativity.

Each sector possesses its own curvature dynamics.

Cosmological Term

$$\Lambda_{(J)}$$

represents the sectoral cosmological constant.

Different sectors may possess distinct vacuum structures.

Entropic Kinetic Term

$$\kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu S \nabla_\nu S.$$

This term describes the propagation of entropy.

It is analogous to the kinetic term of a scalar field.

Entropy therefore acquires genuine dynamical behavior.

Entropic Potential

$$V(S).$$

This term governs the self-interaction of the entropy field.

Different choices of potential correspond to different physical models.

Entropic Coupling Factor

$$e^{S/k_B}.$$

This exponential factor is one of the defining features of ToE.

It directly couples entropy to geometry.

The larger the entropy, the greater the contribution of a configuration to the action.

8.9 The Complete HMAS Action

Applying the Obidi Addition Rule yields

$$I_{LOA} = \sum_J \int \left(R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu S \nabla_\nu S + V(S) \right) e^{S/k_B} \sqrt{|g^{(J)}|} d^4x.$$

This action governs the entire Hybrid Manifold Architecture System.

8.10 Alternative Compact Representation

Using Obidi notation,

$$I_{LOA} = \sum_J \int \mathcal{L}_{(J)} e^{S/k_B} \sqrt{|g_{(J)}|} d^4x.$$

where

$$\mathcal{L}_{(J)} = R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} (\nabla S)^2 + V(S).$$

This form is often convenient for analytical derivations.

8.11 Variational Principle

The fundamental postulate of ToE is

$$\delta I_{LOA} = 0.$$

Variation is performed independently with respect to:

$$g_{\mu\nu}^{(J)},$$

and

$$S.$$

Thus the action generates:

1. Geometric field equations.
2. Entropic field equations.

simultaneously.

8.12 Variation with Respect to the Metric

The metric variation yields

$$\delta I_{LOA} = 0 \implies G_{\mu\nu}^{(J)} + \Lambda_{(J)} g_{\mu\nu}^{(J)} = 8\pi G_{(J)} T_{\mu\nu}^{ent}.$$

These equations are called the **Obidi Field Equations**.

They generalize Einstein's equations to the multi-sector environment of HMAS.

8.13 Variation with Respect to Entropy

Variation with respect to

$$S$$

produces the entropy equation of motion.

Schematically,

$$\frac{\delta I_{LOA}}{\delta S} = 0.$$

The resulting equation governs the evolution of entropy throughout HMAS.

Its precise form depends upon the chosen potential

$$V(S).$$

8.14 Entropy as Source of Geometry

One of the most important conceptual consequences of the LOA is the emergence of geometry from entropy.

In conventional General Relativity:

$$T_{\mu\nu} \rightarrow G_{\mu\nu}.$$

Matter generates curvature.

In ToE:

$$S \rightarrow T_{\mu\nu}^{ent} \rightarrow G_{\mu\nu}.$$

Entropy generates effective stress-energy.

Effective stress-energy generates curvature.

Curvature generates gravity.

Thus entropy becomes the deeper source of geometry.

8.15 Near-Equilibrium Limit

The theory must recover General Relativity in an appropriate limit.

Assume

$$\nabla_\mu S = 0.$$

Then

$$(\nabla S)^2 = 0.$$

The entropy field becomes effectively constant.

Consequently:

$$e^{S/k_B} = \text{constant}.$$

The informational sectors decouple.

Only the Lorentzian sector remains dynamically active.

The action reduces to

$$I_{LOA} \rightarrow I_{EH}.$$

The Einstein–Hilbert Action is recovered.

8.16 Recovery of Einstein Gravity

Under these conditions,

$$G_{\mu\nu}^{(L)} + \Lambda g_{\mu\nu}^{(L)} = 8\pi G T_{\mu\nu}.$$

Identifying

$$g_{\mu\nu}^{(L)} \rightarrow g_{\mu\nu},$$

one obtains

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

This is precisely Einstein's Field Equation.

General Relativity therefore emerges as the low-entropic-gradient limit of the Theory of Entropicity.

8.17 The Role of the LOA in ToE

The Local Obidi Action occupies a position within ToE analogous to that of the Einstein–Hilbert Action within General Relativity.

It provides:

- the dynamical foundation,
- the variational framework,
- the source of field equations,
- the coupling between entropy and geometry.

Every subsequent physical prediction of the theory originates from this action.

8.18 Conceptual Significance

The Local Obidi Action represents a fundamental shift in physical perspective.

Instead of beginning with geometry and deriving entropy,

ToE begins with entropy and derives geometry.

The action therefore embodies the central philosophical commitment of the theory:

$$\text{Entropy} \longrightarrow \text{Information} \longrightarrow \text{Geometry} \longrightarrow \text{Gravity}.$$

This chain is encoded directly within the structure of the action itself.

8.19 Transition to the Obidi Field Equations

The Local Obidi Action provides the variational foundation of the Theory of Entropicity.

The next task is to perform the variational analysis explicitly and derive the resulting equations of motion.

These equations constitute the dynamical heart of the theory.

The following chapter develops the derivation of the **Obidi Field Equations**, establishes their mathematical structure, and demonstrates how Einstein's equations emerge as a limiting case of the broader entropic framework.